

Calculating wave and scattering operators for Schrödinger operators on a one-dimensional lattice with varying spectral multiplicity

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Abstract

We study scattering theory for a class of matrix-valued Schrödinger operators on a one-dimensional lattice. In our model, the free Hamiltonian is given by a variation of the discrete Laplacian, and we consider a matrix potential satisfying a condition of exponential decay. After studying the resolvent and calculating the Green's function, we calculate the fibration of the scattering operator with respect to the energy.

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Introduction

We study scattering theory for a class of matrix-valued Schrödinger operators on $\mathbb{Z}$ of form

$$H = H_0 + V,$$

where the free Hamiltonian H_0 is an energy shift of the discrete Laplace operator composed with the left multiplication by some invertible matrix A and the potential V is a self-adjoint matrix multiplication operator obeying an exponential growth condition (cf. Section 1 for a detailed definition). This model is widely used in the context of low-energy phenomena in solid-state physics.

The novelty of this model is that, if the matrix A that is not a scalar multiply of a unitary matrix, the dimension of the scattering matrix varies depending on the energy. Such models have recently been of interest, for example, in [APRR24], although there, in a similar model, just the case of a potential V supported in a single point has been studied.

A construction of the corresponding scattering matrix has been done in the case $A = \mathbf{1}$ in [BFGS22] and [BFNS24], and for the general case in [BS25]. Even though these papers establish some properties of the constructed scattering matrices, a formal proof that the matrices defined by them do, in fact, coincide with the fibration of the scattering operator with respect to the energy of the free Hamiltonian has not been provided. The present paper aims to close this gap.

Structure of the paper:

In Section 1, we formally introduce the model and prove existence and completeness of wave operators $\Omega_{\pm}$ and scattering operator $S = \Omega_{-}^{*} \Omega_{+}$ (Proposition 1.1).

After this, we show that the function $\sigma(H_0) \ni E \mapsto S^E$, constructed in [BS25], coincides with the fibration of the scattering operator $S = \Omega_{-}^{*} \Omega_{+}$ with respect to the energy E . Our proof of this result is loosely based on the ideas presented in [RS79, Chap. XI.6]. However, since, contrary to our discrete model, [RS79, Chap. XI.6] considers a model on a continuous space, we had to make some profound adaptations.

In Section 2, we then use the results about the resolvent to define a generalized Fourier transform $\mathcal{F}_0$, using the generalized eigenfunctions of H_0 (Definition 2.1.3), as well generalized Fourier transforms $\mathcal{F}_{\pm}$, using the generalized eigenfunctions of H (Theorem 2.2.5). We then show that

$$\Omega_{\pm} = \mathcal{F}_{\pm}^{*} \mathcal{F}_0$$

(Theorem 3.1) and that

$$\mathcal{F}_0 H_0 \mathcal{F}_0^{*} = \int_{\sigma(H_0)}^{\oplus} E \mathbf{1} dE \quad \text{and} \quad \mathcal{F}_0 S \mathcal{F}_0^{*} = \int_{\sigma(H_0)}^{\oplus} S^E dE$$

(Theorem 2.1.4 and Theorem 3.2(b)), which proves that S^E coincides with the fibration of the scattering operator S with respect to the energy E .

1 The model

Given $L \in \mathbb{N}$ and a invertible normal matrix $A \in \text{GL}_L(\mathbb{C})$, we consider the free Hamiltonian H_0 on $\mathcal{H} := \ell^2(\mathbb{Z}, \mathbb{C}^L)$ defined by

$$(H_0 \phi)(n) := A^{*} \phi(n+1) + A \phi(n-1).$$

We now consider a map

$$V : \mathbb{Z} \rightarrow \mathcal{M}_{L \times L}(\mathbb{C})$$

with

$$V(n) = V(n)^* \quad \forall n \in \mathbb{Z},$$

satisfying the condition

$$\sum_{n \in \mathbb{Z}} \|V(n)\| \tau_A^{|n|} < \infty, \quad (1)$$

where

$$\tau_A := \eta_A + \sqrt{\eta_A^2 - 1} \quad \text{with} \quad \eta_A := \|A\| \cdot \|A^{-1}\|.$$

Then, for $\phi \in \mathcal{H}$, we set

$$(V\phi)(n) := V(n)\phi(n)$$

and consider the interaction Hamiltonian

$$H := H_0 + V.$$

We now show that, in our model, the wave operators do, in fact, exist:

Proposition 1.1. *The wave operators*

$$\Omega_{\pm} := s - \lim_{t \rightarrow \mp \infty} e^{itH} e^{-itH_0}$$

exist and are complete, i.e. partial isometries from $P_{ac}(H_0)\mathcal{H} = \mathcal{H}$ to $P_{ac}(H)\mathcal{H}$.

Proof. The operator $H - H_0 = V$ is of trace class, since

$$\|V\|_1 = \sum_{n \in \mathbb{Z}} \|V(n)\|_1 \leq \sum_{n \in \mathbb{Z}} L \|V(n)\| \leq L \cdot \sum_{n \in \mathbb{Z}} \|V(n)\| \tau_A^{|n|} < \infty. \quad (2)$$

The existence and completeness of the wave operators then follow by the Kato-Rosenblum Theorem ([RS79, Thm. XI.8]). $\square$

By [BS25, Sec. 1], after a suitable conjugation with a unitary operator, we can, without loss of generality, assume that

$$A = \text{diag}(a_1, \dots, a_L) \quad \text{with} \quad \{a_1, \dots, a_L\} = \sigma(A)$$

and

$$a_1 \geq \dots \geq a_L > 0. \quad (3)$$

From now on, for the rest of the paper, we will assume that A is of this form. Given this normal form for the matrix A , we can prove the following proposition:

Proposition 1.2. *The operator H_0 has purely absolutely continuous spectrum and*

$$\sigma(H_0) = [-2\|A\|, 2\|A\|].$$

Proof. We consider the operator h_0 on $\ell^2(\mathbb{Z}, \mathbb{C})$ defined by

$$(h_0\phi)(n) := \phi(n+1) + \phi(n-1).$$

Then

$$H_0 = \bigoplus_{k=1}^L a_k h_0.$$

By [BFGS22, Sec. 1.1.1], we have

$$\sigma(h_0) = \sigma_{\text{ac}}(h_0) = [-2, 2],$$

so

$$\begin{aligned} \sigma(H_0) &= \sigma_{\text{ac}}(H_0) = \sigma\left(\bigoplus_{k=1}^L a_k h_0\right) = \bigcup_{k=1}^L a_k \sigma(h_0) = \bigcup_{k=1}^L a_k [-2, 2] \\ &= \bigcup_{k=1}^L [-2a_k, 2a_k] = [-2a_1, 2a_1] = [-2\|A\|, 2\|A\|], \end{aligned}$$

using (3) in the penultimate step. $\square$

2 Generalized Fourier transforms

Throughout this section, we consider the unitary Fourier transform

$$\mathcal{F} : \ell^2(\mathbb{Z}, \mathbb{C}^L) \rightarrow L^2([-\pi, \pi], \mathbb{C}^L)$$

defined by

$$(\mathcal{F}f)(E) := \sum_{n \in \mathbb{Z}} e^{in\theta} f(n), \quad f \in \ell^1(\mathbb{Z}, \mathbb{C}^L).$$

We will construct variants of this Fourier transform, replacing the basis $(e^{in\theta})_{n \in \mathbb{Z}}$ by $(v_k^{E,\sigma}(n))_{n \in \mathbb{Z}}$ in case of the free Hamiltonian and respectively $(w_{k,\pm}^{E,\sigma}(n))_{n \in \mathbb{Z}}$ in case of the interacting Hamiltonian. We will later use these generalized Fourier transforms to calculate the wave- and scattering operators. We start with the free Hamiltonian:

2.1 A Fourier transform for the free Hamiltonian

For the free Hamiltonian, instead of the Hilbert space

$$L^2([-\pi, \pi], \mathbb{C}^L) \cong \bigoplus_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} L^2(\sigma[0, \pi], \mathbb{C}),$$

the image of our generalized Fourier transform will be the following Hilbert space:

Definition 2.1.1. We consider the Hilbert space

$$\mathcal{K} := \bigoplus_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} L^2([-2a_k, 2a_k], \mathbb{C})$$

endowed with the scalar product

$$\langle f|g \rangle := \int^{\mathcal{K}} \overline{f(E)} g(E) dE := \sum_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{-2a_k}^{2a_k} \overline{f_{k,\sigma}(E)} g_{k,\sigma}(E) dE$$

and define the operator

$$\Phi : L^2([-\pi, \pi], \mathbb{C}^L) \rightarrow \mathcal{K}$$

by

$$(\Phi f)_{k,\sigma}(E) := \sigma f_k \left(\sigma \arccos \left(\frac{E}{2a_k} \right) \right) \cdot \sqrt{\langle e_k | \omega^{E, -} e_k \rangle}^{-1}$$

with arccos being the inverse of the function

$$\cos : [0, \pi] \rightarrow [-1, 1].$$

Lemma 2.1.2. *The map $\Phi : L^2([-\pi, \pi], \mathbb{C}^L) \rightarrow \mathcal{K}$ is unitary.*

Proof. For $k = 1, \dots, L$ and $E \in [-2a_k, 2a_k]$, we have

$$a_k(z_{-,k}(E) + z_{-,k}^{-1}(E)) = E,$$

so

$$\begin{aligned} (2a_k)^2 - E^2 &= a_k^2 \left[4 - \left(z_{-,k}(E) + z_{-,k}^{-1}(E) \right)^2 \right] = a_k^2 \left[4 - \left(z_{-,k}(E)^2 + 2 + z_{-,k}^{-2}(E) \right) \right] \\ &= a_k^2 \left[- \left(z_{-,k}(E)^2 - 2 + z_{-,k}^{-2}(E) \right) \right] = -a_k^2 \left(z_{-,k}(E) - z_{-,k}^{-1}(E) \right)^2 \\ &= \langle e_k | \omega^{E, -} e_k \rangle^2. \end{aligned}$$

Since $\langle e_k | \omega^{E, -} e_k \rangle > 0$ by (??), this yields

$$\frac{d}{dE} \left[\arccos \left(\frac{E}{2a_k} \right) \right] = \frac{1}{2a_k} \frac{-1}{\sqrt{1 - \left(\frac{E}{2a_k} \right)^2}} = \frac{-1}{\sqrt{(2a_k)^2 - E^2}} = \frac{-1}{\langle e_k | \omega^{E, -} e_k \rangle}.$$

Now, for $f, g \in \mathcal{K}$, considering the substitution $\theta = \sigma \arccos \left(\frac{E}{2a_k} \right)$, one has

$$\begin{aligned} \langle \Phi f | \Phi g \rangle &= \int^{\mathcal{K}} \overline{(\Phi f)(E)} (\Phi g)(E) dE \\ &= \sum_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{-2a_k}^{2a_k} \overline{f_k \left(\sigma \arccos \left(\frac{E}{2a_k} \right) \right)} g_k \left(\sigma \arccos \left(\frac{E}{2a_k} \right) \right) \cdot \langle e_k | \omega^{E, -} e_k \rangle^{-1} dE \\ &= \sum_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{\sigma\pi}^0 \overline{f_k(\theta)} g_k(\theta) \cdot (-\sigma) dE = \sum_{k \in \{1, \dots, L\}} \int_{-\pi}^{\pi} \overline{f_k(\theta)} g_k(\theta) dE = \langle f | g \rangle, \end{aligned}$$

which shows that Φ is an isometry. Further, Φ is invertible with its inverse being given by

$$(\Phi^{-1} f)_k(\theta) := f_{k, \text{sgn}(\theta)}(2a_k \cos \theta) \cdot \sqrt{\langle e_k | \omega^{2a_k \cos \theta, -} e_k \rangle}^{-1}$$

which implies that Φ is unitary. $\square$

We now define our generalized Fourier transform, of which we will see that it can be calculated using the generalized eigenfunctions of the free Hamiltonian:

Definition 2.1.3. We define

$$\mathcal{F}_0 := \Phi \circ \mathcal{F} : \ell^2(\mathbb{Z}, \mathbb{C}^L) \rightarrow \mathcal{K}.$$

Theorem 2.1.4. *The operator $\mathcal{F}_0$ satisfies the following properties:*

- (a) $\mathcal{F}_0$ is unitary.
- (b) For every $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$, one has

$$(\mathcal{F}_0 f)_{k,\sigma}(E) = \sum_{n \in \mathbb{Z}} \left\langle v_k^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}, \quad k = 1, \dots, L, \sigma \in \{\pm\}, E \in [-2a_k, 2a_k].$$

- (c) $\mathcal{F}_0 H_0 = M_{\text{Id}} \mathcal{F}_0$.

Proof. (a) The unitarity follows by

$$\mathcal{F}_0 = \Phi \circ \mathcal{F}$$

and the fact that $\mathcal{F}$ and Φ are unitary by Lemma 2.1.2.

- (b) We first notice that, for $k = 1, \dots, L$ and $E \in [-2a_k, 2a_k]$, one has

$$a_k \left[e^{i \arccos\left(\frac{E}{2a_k}\right)} + e^{-i \arccos\left(\frac{E}{2a_k}\right)} \right] = 2a_k \cos\left(\arccos\left(\frac{E}{2a_k}\right)\right) = E.$$

Now, for $\sigma \in \{\pm\}$, we have

$$\sigma \operatorname{Im} e^{\sigma i \arccos\left(\frac{E}{2a_k}\right)} = \sigma \sin\left(\sigma \arccos\left(\frac{E}{2a_k}\right)\right) = \sqrt{1 - \left(\frac{E}{2a_k}\right)^2} \geq 0,$$

which, together with Proposition ??(c), implies

$$z_\sigma\left(\frac{E}{a_k}\right) = e^{-\sigma i \arccos\left(\frac{E}{2a_k}\right)}.$$

For $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$, one then gets

$$\begin{aligned} (\mathcal{F}_0 f)_{k,\sigma}(E) &= \sigma (\mathcal{F} f)_k \left(\sigma \arccos\left(\frac{E}{2a_k}\right) \right) \cdot \sqrt{\langle e_k | \omega^E, -e_k \rangle}^{-1} \\ &= \sum_{n \in \mathbb{Z}} \sigma e^{i n \sigma \arccos\left(\frac{E}{2a_k}\right)} f_k(n) \cdot \sqrt{\langle e_k | \omega^E, -e_k \rangle}^{-1} \\ &= \sum_{n \in \mathbb{Z}} \left\langle \sigma \left(e^{-i \sigma \arccos\left(\frac{E}{2a_k}\right)} \right)^n \cdot \sqrt{\langle e_k | \omega^E, -e_k \rangle}^{-1} e_k \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &= \sum_{n \in \mathbb{Z}} \left\langle \sigma \left(z_\sigma\left(\frac{E}{a_k}\right) \right)^n \cdot \nu^E e_k \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &= \sum_{n \in \mathbb{Z}} \left\langle v_k^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}. \end{aligned}$$

(c) By (b), for $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$, $k = 1, \dots, L$ and $\sigma \in \{\pm\}$, we have

$$\begin{aligned} (\mathcal{F}_0 H_0 f)_{k,\sigma}(E) &= \sum_{n \in \mathbb{Z}} \left\langle v_k^{E,\sigma}(n) \middle| (H_0 f)(n) \right\rangle_{\mathbb{C}^L} = \sum_{n \in \mathbb{Z}} \left\langle \left(H_0 v_k^{E,\sigma} \right)(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &= \sum_{n \in \mathbb{Z}} \left\langle E \cdot v_k^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} = E \cdot (\mathcal{F}_0 f)_{k,\sigma}(E) = (M_{\text{Id}} \mathcal{F}_0 f)_{k,\sigma}(E). \end{aligned}$$

The statement then follows, since $\ell^1(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$ is dense. $\square$

We finish this section by the following technical lemma that will be needed later for the construction of generalized Fourier transforms using eigenfunctions of the interacting Hamiltonian:

Lemma 2.1.5. *Let $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$. For $E \in \overline{\mathbb{C}}_{\pm} \setminus \mathfrak{D}$, we define*

$$f_{\pm}^{E,g} := V \sum_{j \in \mathbb{Z}} G_H^{E,\pm}(\cdot, j) g(j) \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$$

and set

$$F_{\pm}^{E,g} := \mathcal{F}_0 \left(g - f_{\pm}^{E,g} \right) \in \mathcal{K}.$$

Further, let $k \in \{1, \dots, L\}$ and $\sigma \in \{\pm\}$ and let $I \subseteq \mathbb{R} \setminus \mathfrak{D}$ and $J \subseteq (-2a_k, 2a_k)$ be compact subsets. Then, the following hold:

(a) The map

$$\overline{\mathbb{C}}_{\pm} \setminus \mathfrak{D} \ni E \mapsto \left(F_{\pm}^{E,g} \right)_{k,\sigma} \Big|_J \in C(J, \mathbb{C})$$

is continuous.

(b) One has

$$\sup_{\substack{E \in I \\ X \in J}} \left| \left(F_{\pm}^{E \pm i\varepsilon, g} - F_{\pm}^{E, g} \right)_{k,\sigma}(X) \right| \xrightarrow{\varepsilon \downarrow 0} 0.$$

(c) One has

$$\lim_{\delta \downarrow 0} \sup_{\substack{E \in I \cap [-2a_k, 2a_k] \\ X \in [E-\delta, E+\delta]}} \left| \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X) - \left(F_{\pm}^{E,g} \right)_{k,\sigma}(E) \right| = 0.$$

(d) For every compact subset $F \subseteq \overline{\mathbb{C}}_{\pm} \setminus \mathfrak{D}$, one has

$$\sup_{E \in F} \left\| F_{\mp}^{E,g} \right\|_{\mathcal{K}} < \infty.$$

Proof. (a) By Lemma ??(c), the function

$$\overline{\mathbb{C}}_{\pm} \setminus \mathfrak{D} \ni E \mapsto f_{\pm}^{E,g} \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$$

is continuous. Since

$$\left(F_{\pm}^{E,g} \right)_{k,\sigma} \Big|_J = \left(\mathcal{F}_0 \left(g - f_{\pm}^{E,g} \right) \right)_{k,\sigma} \Big|_J = \left(\Phi \mathcal{F} \left(g - f_{\pm}^{E,g} \right) \right)_{k,\sigma} \Big|_J$$

and the Fourier transform

$$\mathcal{F} : \ell^1(\mathbb{Z}, \mathbb{C}^L) \rightarrow C([- \pi, \pi], \mathbb{C}^L)$$

is continuous, it suffices to show that the function

$$C([- \pi, \pi], \mathbb{C}^L) \ni f \mapsto (\Phi f)_{k,\sigma} \big|_J \in C(J, \mathbb{C})$$

is continuous. This follows, since, for $f \in C([- \pi, \pi], \mathbb{C}^L)$, we have

$$\begin{aligned} & \left\| (\Phi f)_{k,\sigma} \big|_J \right\|_\infty \\ &= \sup_{X \in J} \left| \sigma f_k \left(\sigma \arccos \left(\frac{X}{2a_k} \right) \right) \cdot \sqrt{\langle e_k | \omega^{X,-} e_k \rangle}^{-1} \right| \\ &\leq \|f\|_\infty \cdot \sup_{X \in J} \left| \sqrt{\langle e_k | \omega^{X,-} e_k \rangle}^{-1} \right| \end{aligned}$$

and, by the continuity of the function

$$X \mapsto \sqrt{\langle e_k | \omega^{X,-} e_k \rangle}^{-1}$$

and the compactness of J , we have

$$\sup_{X \in J} \left| \sqrt{\langle e_k | \omega^{X,-} e_k \rangle}^{-1} \right| < \infty.$$

(b) We consider the compact set

$$F := I + \sigma i[0, 1].$$

By (a), the function

$$F \ni E \mapsto \left(F_{\pm}^{E,g} \right)_{k,\sigma} \big|_J \in C(J, \mathbb{C})$$

is continuous and therefore, by the Heine–Cantor Theorem, uniformly continuous. This immediately yields

$$\begin{aligned} & \sup_{\substack{E \in I \\ X \in J}} \left| \left(F_{\pm}^{E \pm i\varepsilon, g} - F_{\pm}^{E, g} \right)_{k,\sigma} (X) \right| \\ &= \sup_{E \in I} \left\| \left(F_{\pm}^{E \pm i\varepsilon, g} \right)_{k,\sigma} \big|_J - \left(F_{\pm}^{E, g} \right)_{k,\sigma} \big|_J \right\|_\infty \xrightarrow{\varepsilon \downarrow 0} 0. \end{aligned}$$

(c) Since $I \subseteq \mathbb{R} \setminus \mathfrak{D} \subseteq \mathbb{R} \setminus \{\pm 2a_k\}$, the set $I \cap [-2a_k, 2a_k]$ is a compact subset of $(-2a_k, 2a_k)$ and therefore we can choose $r > 0$ such that

$$J := (I \cap [-2a_k, 2a_k]) + [-r, r] \subseteq (-2a_k, 2a_k).$$

We first notice that the function

$$I \times J \ni (E, X) \mapsto \left(F_{\pm}^{E,g} \right)_{k,\sigma} (X) \in \mathbb{C}$$

is continuous since, by (a), we have

$$\begin{aligned}
& \left| \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X) - \left(F_{\pm}^{E',g} \right)_{k,\sigma}(X') \right| \\
& \leq \left| \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X) - \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X') \right| + \left| \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X') - \left(F_{\pm}^{E',g} \right)_{k,\sigma}(X') \right| \\
& \leq \left| \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X) - \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X') \right| + \left\| \left(F_{\pm}^{E,g} - F_{\pm}^{E',g} \right)_{k,\sigma} \right\|_J \\
& \xrightarrow{(E',X') \rightarrow (E,X)} 0.
\end{aligned}$$

As a continuous function on a compact set, by the Heine–Cantor Theorem, it is uniformly continuous. This implies

$$\lim_{\delta \downarrow 0} \sup_{\substack{(E,X), (E',X') \in I \times J \\ (E-E')^2 + (X-X')^2 \leq \delta^2}} \left| \left(F_{\pm}^{E,g} \right)_{k,\sigma}(X) - \left(F_{\pm}^{E',g} \right)_{k,\sigma}(X') \right| = 0.$$

The statement then follows, since

$$\begin{aligned}
& \{((E, X), (E, E)) : E \in I \cap [-2a_k, 2a_k], X \in [E - \delta, E + \delta]\} \\
& \subseteq \{((E, X), (E', X')) \in (I \times J)^2 : (E - E')^2 + (X - X')^2 \leq \delta^2\}.
\end{aligned}$$

(d) By Lemma ??(c), the function

$$\overline{\mathbb{C}_{\pm}} \setminus \mathfrak{D} \ni E \mapsto f_{\pm}^{E,g} \in \ell^1(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$$

is continuous. Since $F_{\pm}^{E,g} = \mathcal{F}_0(g - f_{\pm}^{E,g})$, the statement immediately follows by the continuity of $\mathcal{F}_0$ due to Theorem 2.1.4(a). $\square$

2.2 A Fourier transform for the interacting Hamiltonian

In this section, we want to define generalized Fourier transforms $\mathcal{F}_{\pm}$ analogous to $\mathcal{F}_0$ in a way such that Theorem 2.1.4(b) holds for the eigenfunctions of the interacting Hamiltonian instead of the free Hamiltonian, i.e.

$$(\mathcal{F}_{\pm} f)_{k,\sigma}(E) := \sum_{n \in \mathbb{Z}} \left\langle w_{k,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}, \quad f \in \ell^1(\mathbb{Z}, \mathbb{C}^L).$$

The difficulty here is to see that, defined in this way, $\mathcal{F}_{\pm} f$ is an element of $\mathcal{K}$ and that $\mathcal{F}_{\pm}$ can be extended to a unitary operator on $\ell^2(\mathbb{Z}, \mathbb{C}^L)$. We will address these issues later in the section. We start with the following definition:

Definition 2.2.1. Let $I \subseteq \mathbb{R}$. We write $\mathcal{M}(I, \mathbb{C})$ for the set of equivalence classes of Borel measurable functions up to values on a Lebesgue zero-set. Further, we define

$$\tilde{\mathcal{F}}_{\pm} : \ell^1(\mathbb{Z}, \mathbb{C}^L) \rightarrow \bigoplus_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \mathcal{M}([-2a_k, 2a_k], \mathbb{C})$$

by

$$(\tilde{\mathcal{F}}_{\pm} f)_{k,\sigma}(E) := \sum_{n \in \mathbb{Z}} \left\langle w_{k,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}, \quad f \in \ell^1(\mathbb{Z}, \mathbb{C}^L).$$

We now link the so defined operators $\tilde{\mathcal{F}}_{\pm}$ to Lemma 2.1.5:

Lemma 2.2.2. *Let $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, $k \in \{1, \dots, L\}$ and $\sigma \in \{\pm\}$. Then, for every $E \in [-2a_k, 2a_k] \setminus \mathfrak{D}$, one has*

$$\left(F_{\mp}^{E,g}\right)_{k,\sigma}(E) = (\tilde{\mathcal{F}}_{\pm}g)_{k,\sigma}(E).$$

Proof. By Lemma ??(c) and Theorem 2.1.4(a), we get

$$\left(\mathcal{F}_0 f_{\mp}^{E,g}\right)_{k,\sigma}(E) = \lim_{\varepsilon \downarrow 0} \left(\mathcal{F}_0 f_{\mp}^{E \mp i\varepsilon, g}\right)_{k,\sigma}(E).$$

Using that, by Theorem ??(g), we have

$$f_{\mp}^{E \mp i\varepsilon, g} = V(H - E \pm i\varepsilon)^{-1}g,$$

this yields

$$\begin{aligned} \left(\mathcal{F}_0 f_{\mp}^{E,g}\right)_{k,\sigma}(E) &= \lim_{\varepsilon \downarrow 0} \left(\mathcal{F}_0 V(H - E \pm i\varepsilon)^{-1}g\right)_{k,\sigma}(E) \\ &= \lim_{\varepsilon \downarrow 0} \sum_{n \in \mathbb{Z}} \left\langle v_k^{E,\sigma}(n) \left| (V(H - E \pm i\varepsilon)^{-1}g)(n) \right\rangle_{\mathbb{C}^L} \\ &= \lim_{\varepsilon \downarrow 0} \sum_{n \in \mathbb{Z}} \left\langle \left((H - E \mp i\varepsilon)^{-1} V v_k^{E,\sigma} \right)(n) \left| g(n) \right\rangle_{\mathbb{C}^L} \right. \\ &= \lim_{\varepsilon \downarrow 0} \sum_{n \in \mathbb{Z}} \left\langle v_k^{E,\sigma}(n) - w_{k,\pm}^{E,\sigma}(n) \left| g(n) \right\rangle_{\mathbb{C}^L} \right. \\ &= ((\mathcal{F}_0 - \tilde{\mathcal{F}}_{\pm})g)_{k,\sigma}(E), \end{aligned}$$

using Theorem ?? in the penultimate and Theorem 2.1.4(b) in the last step. We then get

$$\begin{aligned} (\tilde{\mathcal{F}}_{\pm}g)_{k,\sigma}(E) &= (\mathcal{F}_0 g - (\mathcal{F}_0 - \tilde{\mathcal{F}}_{\pm})g)_{k,\sigma}(E) \\ &= \left(\mathcal{F}_0 g - \mathcal{F}_0 f_{\mp}^{E,g}\right)_{k,\sigma}(E) = \left(F_{\mp}^{E,g}\right)_{k,\sigma}(E). \quad \square \end{aligned}$$

Proposition 2.2.3. *Let $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$ and let $I \subseteq \mathbb{R} \setminus \mathfrak{D}$ by a compact subset. Then, the following hold:*

(a) *For $k \in \{1, \dots, L\}$ and $\sigma \in \{\pm\}$, one has*

$$\begin{aligned} &\frac{1}{\pi} \int_I \int_{[-2a_k, 2a_k]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \overline{\left(F_{\mp}^{E \mp i\varepsilon, f}\right)_{k,\sigma}(X)} \left(F_{\mp}^{E \mp i\varepsilon, g}\right)_{k,\sigma}(X) dX dE \\ &\xrightarrow{\varepsilon \downarrow 0} \int_{[-2a_k, 2a_k]} \chi_I(E) \overline{\left(\tilde{\mathcal{F}}_{\pm}f\right)_{k,\sigma}(E)} \left(\tilde{\mathcal{F}}_{\pm}g\right)_{k,\sigma}(E) dE. \end{aligned}$$

(b) *One has*

$$\begin{aligned} &\frac{\varepsilon}{\pi} \int_I \left\langle (H - E \pm i\varepsilon)^{-1}f \left| (H - E \pm i\varepsilon)^{-1}g \right\rangle dE \\ &\xrightarrow{\varepsilon \downarrow 0} \int^{\mathcal{K}} \chi_I(E) \overline{\left(\tilde{\mathcal{F}}_{\pm}f\right)(E)} \left(\tilde{\mathcal{F}}_{\pm}g\right)(E) dE. \end{aligned}$$

Proof. (a) Set

$$h(Y, X) := \overline{\left(F_{\mp}^{Y,f}\right)_{k,\sigma}}(X) \left(F_{\mp}^{Y,g}\right)_{k,\sigma}(X).$$

By Lemma 2.2.2, we have

$$h(E, E) = \overline{\left(F_{\mp}^{E,f}\right)_{k,\sigma}}(E) \left(F_{\mp}^{E,g}\right)_{k,\sigma}(E) = \overline{\left(\tilde{\mathcal{F}}_{\pm} f\right)_{k,\sigma}}(E) \left(\tilde{\mathcal{F}}_{\pm} g\right)_{k,\sigma}(E),$$

so, setting

$$\mathcal{I} := \int_{I \cap [-2a_k, 2a_k]} h(E, E) dE = \int_{[-2a_k, 2a_k]} \chi_I(E) h(E, E) dE$$

we have to show that

$$\frac{1}{\pi} \int_I \int_{[-2a_k, 2a_k]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE \xrightarrow{\varepsilon \downarrow 0} \mathcal{I}. \quad (4)$$

Since $I \subseteq \mathbb{R} \setminus \mathfrak{D} \subseteq \mathbb{R} \setminus \{\pm 2a_k\}$, the set $I \cap [-2a_k, 2a_k]$ is a compact subset of $(-2a_k, 2a_k)$ and therefore, for $\delta > 0$ sufficiently small, one has

$$(I \cap [-2a_k, 2a_k]) + [-\delta, \delta] \subseteq (-2a_k, 2a_k).$$

We then set

$$M_\delta := \{(E, X) \in I \times [-2a_k, 2a_k] : E \in I \cap [-2a_k, 2a_k], X \in [E - \delta, E + \delta]\}$$

and

$$N_\delta := (I \times [-2a_k, 2a_k]) \setminus M_\delta.$$

We write

$$\frac{1}{\pi} \int_I \int_{[-2a_k, 2a_k]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE = A_\delta(\varepsilon) + B_\delta(\varepsilon) + C_\delta(\varepsilon) + D_\delta(\varepsilon),$$

with

$$\begin{aligned} A_\delta(\varepsilon) &:= \frac{1}{\pi} \int_{N_\delta} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE, \\ B_\delta(\varepsilon) &:= \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} [h(E \mp i\varepsilon, X) - h(E, X)] dX dE, \\ C_\delta(\varepsilon) &:= \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} [h(E, X) - h(E, E)] dX dE, \\ D_\delta(\varepsilon) &:= \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E, E) dX dE. \end{aligned}$$

We will now prove (4), by showing the following:

- (1) For every $\delta > 0$, one has $\lim_{\varepsilon \downarrow 0} A_\delta(\varepsilon) = 0$.
- (2) For every $\delta > 0$, one has $\lim_{\varepsilon \downarrow 0} B_\delta(\varepsilon) = 0$.

(3) One has $\limsup_{\delta \downarrow 0} \limsup_{\varepsilon > 0} |C_\delta(\varepsilon)| = 0$.

(4) For every $\delta > 0$, one has $\lim_{\varepsilon \downarrow 0} D_\delta(\varepsilon) = \mathcal{I}$.

The statement then follows since, for every $\delta > 0$, one has

$$\begin{aligned}
& \limsup_{\varepsilon \downarrow 0} \left| \frac{1}{\pi} \int_I \int_{[-2a_k, 2a_k]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE - \mathcal{I} \right| \\
&= \limsup_{\varepsilon \downarrow 0} |A_\delta(\varepsilon) + B_\delta(\varepsilon) + C_\delta(\varepsilon) + D_\delta(\varepsilon) - \mathcal{I}| \\
&\leq \limsup_{\varepsilon \downarrow 0} |A_\delta(\varepsilon)| + \limsup_{\varepsilon \downarrow 0} |B_\delta(\varepsilon)| + \limsup_{\varepsilon \downarrow 0} |C_\delta(\varepsilon)| + \limsup_{\varepsilon \downarrow 0} |D_\delta(\varepsilon) - \mathcal{I}| \\
&= \limsup_{\varepsilon \downarrow 0} |C_\delta(\varepsilon)| \leq \sup_{\varepsilon > 0} |C_\delta(\varepsilon)|
\end{aligned}$$

and therefore

$$\begin{aligned}
& \limsup_{\varepsilon \downarrow 0} \left| \frac{1}{\pi} \int_I \int_{[-2a_k, 2a_k]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE - \mathcal{I} \right| \\
&\leq \inf_{\delta > 0} \sup_{\varepsilon > 0} |C_\delta(\varepsilon)| \leq \limsup_{\delta \downarrow 0} \sup_{\varepsilon > 0} |C_\delta(\varepsilon)| = 0.
\end{aligned}$$

We now prove the above statements:

(1) Since $I \subseteq \mathbb{R} \setminus \mathfrak{D} \subseteq \mathbb{R} \setminus \{\pm 2a_k\}$, the set $I \setminus [-2a_k, 2a_k]$ is a compact subset of $\mathbb{R} \setminus [-2a_k, 2a_k]$ and therefore

$$r := \text{dist}(I \setminus [-2a_k, 2a_k], [-2a_k, 2a_k]) > 0.$$

For $(E, X) \in N_\delta$, we either have $E \in I \setminus [-2a_k, 2a_k]$ or $E \in I \cap [-2a_k, 2a_k]$.

In the first case, we get

$$\frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \leq \frac{\varepsilon}{r^2}$$

and in the second case, we have $X \notin [E - \delta, E + \delta]$, so

$$\frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \leq \frac{\varepsilon}{\delta^2}.$$

Therefore, setting $C := \pi^{-1} \max\{r^{-2}, \delta^{-2}\}$, we get

$$\frac{1}{\pi} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \leq C\varepsilon \quad \forall (E, X) \in N_\delta.$$

For $\varepsilon \leq 1$, this implies

$$\begin{aligned}
|A_\delta(\varepsilon)| &= \left| \frac{1}{\pi} \int_{N_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE \right| \\
&\leq C\varepsilon \int_{N_\delta} |h(E \mp i\varepsilon, X)| dX dE \\
&\leq C\varepsilon \int_I \int_{[-2a_k, 2a_k]} |h(E \mp i\varepsilon, X)| dX dE \\
&= C\varepsilon \int_I \int_{[-2a_k, 2a_k]} \left| \overline{\left(F_{\mp}^{E \mp i\varepsilon, f} \right)_{k, \sigma}(X)} \left(F_{\mp}^{E \mp i\varepsilon, g} \right)_{k, \sigma}(X) \right| dX dE \\
&\leq C\varepsilon \int_I \left\| F_{\mp}^{E \mp i\varepsilon, f} \right\|_{\mathcal{K}} \cdot \left\| F_{\mp}^{E \mp i\varepsilon, g} \right\|_{\mathcal{K}} dE \\
&\leq C\varepsilon \cdot \lambda_1(I) \cdot \sup_{Y \in I \mp i[0, 1]} \left\| F_{\mp}^{Y, f} \right\|_{\mathcal{K}} \cdot \left\| F_{\mp}^{Y, g} \right\|_{\mathcal{K}} \xrightarrow{\varepsilon \downarrow 0} 0,
\end{aligned}$$

denoting by λ_1 the Lebesgue measure on $\mathbb{R}$ and using that, by Lemma 2.1.5(d), we have

$$\sup_{Y \in I \mp i[0, 1]} \left\| F_{\mp}^{Y, f} \right\|_{\mathcal{K}} \cdot \left\| F_{\mp}^{Y, g} \right\|_{\mathcal{K}} < \infty.$$

(2) Setting

$$J := (I \cap [-2a_k, 2a_k]) + [-\delta, \delta]$$

and defining

$$\begin{aligned}
S(\varepsilon) &:= \sup_{\substack{E \in I \cap [-2a_k, 2a_k] \\ X \in [E-\delta, E+\delta]}} |h(E \mp i\varepsilon, X) - h(E, X)| \\
&\leq \sup_{\substack{E \in I \cap [-2a_k, 2a_k] \\ X \in J}} |h(E \mp i\varepsilon, X) - h(E, X)|,
\end{aligned}$$

by Lemma 2.1.5(b), we have

$$\lim_{\varepsilon \downarrow 0} S(\varepsilon) = 0.$$

Using that

$$\begin{aligned}
\int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX &= \int_{-\delta}^{\delta} \frac{\varepsilon}{X^2 + \varepsilon^2} dX \\
&= \int_{-\delta\varepsilon^{-1}}^{\delta\varepsilon^{-1}} \frac{1}{X^2 + 1} dX \leq \pi,
\end{aligned} \tag{5}$$

we then get

$$\begin{aligned}
|B_\delta(\varepsilon)| &= \left| \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} [h(E \mp i\varepsilon, X) - h(E, X)] dX dE \right| \\
&= \left| \frac{1}{\pi} \int_{I \cap [-2a_k, 2a_k]} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} [h(E \mp i\varepsilon, X) - h(E, X)] dX dE \right| \\
&\leq S(\varepsilon) \cdot \int_{I \cap [-2a_k, 2a_k]} \frac{1}{\pi} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&\leq S(\varepsilon) \cdot \lambda_1(I) \xrightarrow{\varepsilon \downarrow 0} 0.
\end{aligned}$$

(3) Setting

$$T(\delta) := \sup_{(E, X) \in M_\delta} |h(E, X) - h(E, E)|,$$

by Lemma 2.1.5(c), we have

$$\lim_{\delta \downarrow 0} T(\delta) = 0.$$

With (5), this yields

$$\begin{aligned}
\sup_{\varepsilon > 0} |C_\delta(\varepsilon)| &= \sup_{\varepsilon > 0} \left| \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} [h(E, X) - h(E, E)] dX dE \right| \\
&\leq T(\delta) \cdot \sup_{\varepsilon > 0} \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&= T(\delta) \cdot \sup_{\varepsilon > 0} \frac{1}{\pi} \int_{I \cap [-2a_k, 2a_k]} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&\leq T(\delta) \cdot \lambda_1(I) \xrightarrow{\delta \downarrow 0} 0.
\end{aligned}$$

(4) Using (5), we have

$$\begin{aligned}
D_\delta(\varepsilon) &= \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} h(E, E) dX dE \\
&= \frac{1}{\pi} \int_{I \cap [-2a_k, 2a_k]} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} h(E, E) dX dE \\
&= \frac{1}{\pi} \int_{I \cap [-2a_k, 2a_k]} h(E, E) \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&= \frac{1}{\pi} \int_{I \cap [-2a_k, 2a_k]} h(E, E) \int_{-\delta\varepsilon^{-1}}^{\delta\varepsilon^{-1}} \frac{1}{X^2 + 1} dX dE \\
&= \frac{1}{\pi} \int_{-\delta\varepsilon^{-1}}^{\delta\varepsilon^{-1}} \frac{1}{X^2 + 1} dX \int_{I \cap [-2a_k, 2a_k]} h(E, E) dE \\
&\xrightarrow{\varepsilon \downarrow 0} \int_{I \cap [-2a_k, 2a_k]} h(E, E) dE = \mathcal{I}.
\end{aligned}$$

(b) By summing statement (a) over $k \in \{1, \dots, L\}$ and $\sigma \in \{\pm\}$, we immediately get

$$\begin{aligned} & \frac{1}{\pi} \int_I \int^\mathcal{K} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} \overline{F_\mp^{E \mp i\varepsilon, f}(X)} F_\mp^{E \mp i\varepsilon, g}(X) dX dE \\ & \xrightarrow{\varepsilon \downarrow 0} \int^\mathcal{K} \chi_I(E) \overline{\tilde{\mathcal{F}}_\pm f(E)} \tilde{\mathcal{F}}_\pm g(E) dE. \end{aligned} \quad (6)$$

By Theorem ??(g), we have

$$\begin{aligned} F_\mp^{E \mp i\varepsilon, g} &= \mathcal{F}_0 \left(g - f_\mp^{E \mp i\varepsilon, g} \right) = \mathcal{F}_0 \left(g - V(H - E \pm i\varepsilon)^{-1} g \right) \\ &= \mathcal{F}_0 \left([(H - E \pm i\varepsilon) - V] (H - E \pm i\varepsilon)^{-1} g \right) \\ &= \mathcal{F}_0 \left((H_0 - E \pm i\varepsilon) (H - E \mp i\varepsilon)^{-1} g \right), \end{aligned}$$

so, by Theorem 2.1.4(c), we get

$$\begin{aligned} F_\mp^{E \mp i\varepsilon, g}(X) &= \mathcal{F}_0 \left((H_0 - E \pm i\varepsilon) (H - E \pm i\varepsilon)^{-1} g \right) (X) \\ &= (X - E \pm i\varepsilon) \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X). \end{aligned}$$

This yields

$$\begin{aligned} & \overline{F_\mp^{E \mp i\varepsilon, f}(X)} F_\mp^{E \mp i\varepsilon, g}(X) \\ &= \overline{(X - E \pm i\varepsilon) \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} f \right) (X)} (X - E \pm i\varepsilon) \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X) \\ &= [(X - E)^2 + \varepsilon^2] \cdot \overline{\mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} f \right) (X)} \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X) \end{aligned}$$

and therefore

$$\begin{aligned} & \frac{1}{\pi} \int_I \int^\mathcal{K} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} \overline{F_\mp^{E \mp i\varepsilon, f}(X)} F_\mp^{E \mp i\varepsilon, g}(X) dX dE \\ &= \frac{\varepsilon}{\pi} \int_I \int^\mathcal{K} \overline{\mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} f \right) (X)} \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X) dX dE \\ &= \frac{\varepsilon}{\pi} \int_I \langle \mathcal{F}_0 (H - E \pm i\varepsilon)^{-1} f | \mathcal{F}_0 (H - E \pm i\varepsilon)^{-1} g \rangle dE \\ &= \frac{\varepsilon}{\pi} \int_I \langle (H - E \pm i\varepsilon)^{-1} f | (H - E \pm i\varepsilon)^{-1} g \rangle dE, \end{aligned}$$

using in the last step that, by Theorem 2.1.4(a), the operator $\mathcal{F}_0$ is unitary. Together with (6), this implies the statement. $\square$

The above limit will be the key ingredient in proving the following result:

Proposition 2.2.4. *The following hold:*

(a) *One has*

$$\tilde{\mathcal{F}}_\pm(c_{00}(\mathbb{Z}, \mathbb{C}^L)) \subseteq \mathcal{K}.$$

(b) *The operator H has no singular continuous spectrum.*

(c) For every $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, one has

$$\langle f | P_{\text{ac}}(H) g \rangle = \left\langle \tilde{\mathcal{F}}_{\pm} f \middle| \tilde{\mathcal{F}}_{\pm} g \right\rangle_{\mathcal{K}}.$$

Proof. (a) For $E \in \mathbb{R}$ and $\varepsilon > 0$, one has

$$\begin{aligned} (H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1} &= 2i\varepsilon(H - E - i\varepsilon)^{-1}(H - E + i\varepsilon)^{-1}, \\ &= 2i\varepsilon(H - E + i\varepsilon)^{-1}(H - E - i\varepsilon)^{-1}, \end{aligned}$$

so, by Proposition 2.2.3(b), for every compact subset $I \subseteq \mathbb{R} \setminus \mathfrak{D}$ and all functions $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we get

$$\begin{aligned} & \frac{1}{2\pi i} \int_I \langle f | [(H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1}] g \rangle dE \\ &= \frac{1}{2\pi i} \int_I \langle f | 2i\varepsilon(H - E \mp i\varepsilon)^{-1}(H - E \pm i\varepsilon)^{-1} g \rangle dE \\ &= \frac{\varepsilon}{\pi} \int_I \langle (H - E \pm i\varepsilon)^{-1} f | (H - E \pm i\varepsilon)^{-1} g \rangle dE \\ &\xrightarrow{\varepsilon \downarrow 0} \int^{\mathcal{K}} \chi_I(E) \overline{(\tilde{\mathcal{F}}_{\pm} f)}(E) (\tilde{\mathcal{F}}_{\pm} g)(E) dE. \end{aligned} \tag{7}$$

On the other hand, for every compact interval $[a, b]$, by Stone's formula, one has

$$\begin{aligned} & \left\langle f \middle| \frac{1}{2} [P_{[a,b]}(H) + P_{(a,b)}(H)] g \right\rangle \\ &= \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_a^b \langle f | [(H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1}] g \rangle dE. \end{aligned}$$

If $[a, b] \cap \sigma_{\text{pp}}(H) = \emptyset$, we have

$$\frac{1}{2} [P_{[a,b]}(H) + P_{(a,b)}(H)] = P_{[a,b]}(H),$$

so, for $[a, b] \subseteq \mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$, we get

$$\begin{aligned} \langle f | P_{[a,b]}(H) g \rangle &= \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{[a,b]} \langle f | [(H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1}] g \rangle dE \\ &= \int^{\mathcal{K}} \chi_{[a,b]}(E) \overline{(\tilde{\mathcal{F}}_{\pm} f)}(E) (\tilde{\mathcal{F}}_{\pm} g)(E) dE, \end{aligned} \tag{8}$$

using (7) in the last step.

Now, for every compact subset $I \subseteq \mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$, there exist compact intervals $(I_n)_{n \in \mathbb{N}}$ in $\mathbb{R} \setminus \mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$, such that $I = \bigcup_{n \in \mathbb{N}} I_n$. Then, by Equation (8) and

the σ -additivity of the spectral measure of H , for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we get

$$\begin{aligned} \langle f | P_I(H) f \rangle &= \sum_{n \in \mathbb{N}} \langle f | P_{I_n}(H) f \rangle = \sum_{n \in \mathbb{N}} \int^{\mathcal{K}} \chi_{I_n}(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE \\ &= \lim_{N \rightarrow \infty} \int^{\mathcal{K}} \sum_{n=1}^N \chi_{I_n}(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE \\ &= \int^{\mathcal{K}} \chi_I(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE, \end{aligned} \quad (9)$$

using the Monotone Convergence Theorem in the last step.

Now, for every open subset $O \subseteq \mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$, setting

$$O_n := \left\{ E \in O : \text{dist}(E, O^c) \geq \frac{1}{n}, |E| \leq n \right\}, \quad n \in \mathbb{N},$$

$(O_n)_{n \in \mathbb{N}}$ is an increasing sequence of compact subsets of $\mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$ with

$$\bigcup_{n \in \mathbb{N}} O_n = O$$

and therefore, by Equation (9) and the σ -additivity of the spectral measure of H , for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we get

$$\begin{aligned} \langle f | P_O(H) f \rangle &= \lim_{n \rightarrow \infty} \langle f | P_{O_n}(H) f \rangle = \lim_{n \rightarrow \infty} \int^{\mathcal{K}} \chi_{O_n}(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE \\ &= \int^{\mathcal{K}} \chi_O(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE, \end{aligned} \quad (10)$$

using the Monotone Convergence Theorem in the last step.

Since, by Proposition ??, the set $\mathfrak{D}' := \overline{\mathfrak{D} \cup \sigma_{\text{pp}}(H)} \supseteq \sigma_{\text{pp}}(H)$ is at most countable, we have

$$P_{\mathfrak{D}'}(H) = P_{\text{pp}}(H) \quad (11)$$

and therefore

$$P_{\mathbb{R} \setminus \mathfrak{D}'}(H) = P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H).$$

For the open set $O = \mathbb{R} \setminus \mathfrak{D}'$ and $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, using Equation (10), this yields

$$\begin{aligned} \langle f | P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H) f \rangle &= \langle f | P_{\mathbb{R} \setminus \mathfrak{D}'}(H) f \rangle \\ &= \int^{\mathcal{K}} \chi_{\mathbb{R} \setminus \mathfrak{D}'}(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE \\ &= \int^{\mathcal{K}} \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE = \left\| \tilde{\mathcal{F}}_{\pm} f \right\|_{\mathcal{K}}^2, \end{aligned} \quad (12)$$

using in the last step that $\mathfrak{D}'$ is at most countable and therefore a Lebesgue zero-set. This immediately implies $\tilde{\mathcal{F}}_{\pm} f \in \mathcal{K}$.

- (b) Given a Lebesgue zero-set $C \subseteq \mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$, by the outer regularity of the Lebesgue measure, there exists a decreasing sequence of open subsets $(C_n)_{n \in \mathbb{N}}$ of $\mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$ with $C \subseteq C_n$, such that

$$\lim_{n \rightarrow \infty} \lambda_1(C_n) = \lambda_1(C) = 0.$$

Then, by the Monotone Convergence Theorem and Equation (10), for every element $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we get

$$\begin{aligned} 0 &\leq \langle f | P_C(H) f \rangle \leq \lim_{n \rightarrow \infty} \langle f | P_{C_n}(H) f \rangle \\ &= \lim_{n \rightarrow \infty} \int_{\mathcal{K}} \chi_{C_n}(E) \overline{(\mathcal{F}_{\pm} f)(E)} (\mathcal{F}_{\pm} f)(E) dE = 0 \end{aligned}$$

and therefore $P_C(H) = 0$ since $c_{00}(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$ is dense. This proves that H has purely absolutely continuous spectrum on $\mathbb{R} \setminus (\mathfrak{D} \cup \sigma_{\text{pp}}(H))$. Together with (11), this implies the absence of singular continuous spectrum for the operator H .

- (c) By (b), we have

$$P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H) = P_{\text{ac}}(H),$$

which together with Equation (12) yields

$$\langle f | P_{\text{ac}}(H) f \rangle = \langle f | P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H) f \rangle = \left\| \tilde{\mathcal{F}}_{\pm} f \right\|_{\mathcal{K}}^2 = \left\langle \tilde{\mathcal{F}}_{\pm} f \middle| \tilde{\mathcal{F}}_{\pm} f \right\rangle_{\mathcal{K}}$$

for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$. By polarization, this implies

$$\langle f | P_{\text{ac}}(H) g \rangle = \left\langle \tilde{\mathcal{F}}_{\pm} f \middle| \tilde{\mathcal{F}}_{\pm} g \right\rangle_{\mathcal{K}} \quad \forall f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L). \quad \square$$

With this preparation, adapting the ideas in the proof of [AW21, Thm. 4.3.1] to our model, we can finally show that, similar to $\mathcal{F}_0$, the operators $\tilde{\mathcal{F}}_{\pm}$ give rise to unitary operators from $\ell^2(\mathbb{Z}, \mathbb{C})$ to $\mathcal{K}$:

Theorem 2.2.5. *The maps $\tilde{\mathcal{F}}_{\pm}|_{\mathcal{K}}$ extend to bounded linear operators*

$$\mathcal{F}_{\pm} : \ell^2(\mathbb{Z}, \mathbb{C}) \rightarrow \mathcal{K}$$

such that the following hold:

- (a) $\mathcal{F}_{\pm}^* \mathcal{F}_{\pm} = P_{\text{ac}}(H)$.
- (b) $\mathcal{F}_{\pm} H = M_{\text{Id}} \mathcal{F}_{\pm}$.
- (c) $\mathcal{F}_{\pm} \mathcal{F}_{\pm}^* = \mathbf{1}_{\mathcal{K}}$.

Proof. By Proposition 2.2.4, for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we have

$$\left\| \tilde{\mathcal{F}}_{\pm} f \right\|_{\mathcal{K}}^2 = \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} f \rangle_{\mathcal{K}} = \langle f | P_{\text{ac}}(H) f \rangle = \|P_{\text{ac}}(H) f\|_{\mathcal{K}}^2 \leq \|f\|_{\mathcal{K}}^2,$$

which shows that $\tilde{\mathcal{F}}_{\pm}|_{c_{00}(\mathbb{Z}, \mathbb{C}^L)}$ extends to a linear operator

$$\mathcal{F}_{\pm} : \ell^2(\mathbb{Z}, \mathbb{C}) \rightarrow \mathcal{K}$$

with $\|\mathcal{F}_\pm\| \leq 1$. To show that $\mathcal{F}_\pm$ is an extension of $\tilde{\mathcal{F}}_\pm|^\mathcal{K}$, it remains to show that

$$\mathcal{F}_\pm f = \tilde{\mathcal{F}}_\pm f \quad (13)$$

for every $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$. By construction, Equation (13) holds for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$. Now, let $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$ and consider the sequence $(f_N)_{N \in \mathbb{N}} \subseteq c_{00}(\mathbb{Z}, \mathbb{C}^L)$ defined by

$$f_N(n) = \begin{cases} f(n) & \text{if } n \leq N \\ 0 & \text{if } n > N. \end{cases}$$

Then, for every $E \in [-2a_k, 2a_k] \setminus \mathfrak{D}$, we have

$$\begin{aligned} \left(\tilde{\mathcal{F}}_\pm f_N \right)_{k,\sigma}(E) &= \sum_{n \in \mathbb{Z}} \left\langle w_{k,\pm}^{E,\sigma}(n) \middle| f_N(n) \right\rangle_{\mathbb{C}^L} = \sum_{n=1}^N \left\langle w_{k,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &\xrightarrow{N \rightarrow \infty} \sum_{n \in \mathbb{Z}} \left\langle w_{k,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} = \left(\tilde{\mathcal{F}}_\pm f \right)_{k,\sigma}(E). \end{aligned} \quad (14)$$

On the other hand, one has $f_N \xrightarrow{N \rightarrow \infty} f$ in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$ and therefore, since $\mathcal{F}_\pm$ is continuous, we also have $\mathcal{F}_\pm f_N \xrightarrow{N \rightarrow \infty} \mathcal{F}_\pm f$ in $\mathcal{K}$. For $k \in \{1, \dots, L\}$ and $\sigma \in \{\pm\}$, this implies the existence of a subsequence $(f_{N_j})_{j \in \mathbb{N}}$ such that

$$(\mathcal{F}_\pm f_{N_j})_{k,\sigma}(E) \xrightarrow{j \rightarrow \infty} (\mathcal{F}_\pm f)_{k,\sigma}(E)$$

for almost every $E \in [-2a_k, 2a_k]$. Together with (14), this implies that, for almost every $E \in [-2a_k, 2a_k]$, one has

$$(\mathcal{F}_\pm f)_{k,\sigma}(E) = \lim_{j \rightarrow \infty} (\mathcal{F}_\pm f_{N_j})_{k,\sigma}(E) = \lim_{j \rightarrow \infty} \left(\tilde{\mathcal{F}}_\pm f_{N_j} \right)_{k,\sigma}(E) = \left(\tilde{\mathcal{F}}_\pm f \right)_{k,\sigma}(E)$$

and therefore

$$\mathcal{F}_\pm f = \tilde{\mathcal{F}}_\pm f.$$

We now show that the maps $\mathcal{F}_\pm$ satisfy the above statements:

(a) By Proposition 2.2.4, for every $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we have

$$\langle f | \mathcal{F}_\pm^* \mathcal{F}_\pm g \rangle_{\mathcal{K}} = \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle_{\mathcal{K}} = \left\langle \tilde{\mathcal{F}}_\pm f \middle| \tilde{\mathcal{F}}_\pm g \right\rangle_{\mathcal{K}} = \langle f | P_{\text{ac}}(H) g \rangle,$$

which implies the statement since $c_{00}(\mathbb{Z}, \mathbb{C}^L)$ is dense in $\ell^2(\mathbb{Z}, \mathbb{C})$.

(b) This is proven analogously to Theorem 2.1.4(c).

(c) By (a), for $f, g \in \ell^2(\mathbb{Z}, \mathbb{C})$, we have

$$\begin{aligned} \langle \mathcal{F}_\pm f | (\mathcal{F}_\pm \mathcal{F}_\pm^*) \mathcal{F}_\pm g \rangle &= \langle f | (\mathcal{F}_\pm^* \mathcal{F}_\pm) (\mathcal{F}_\pm^* \mathcal{F}_\pm) g \rangle = \langle f | P_{\text{ac}}(H) P_{\text{ac}}(H) g \rangle \\ &= \langle f | P_{\text{ac}}(H) g \rangle = \langle f | \mathcal{F}_\pm^* \mathcal{F}_\pm g \rangle = \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle = \langle \mathcal{F}_\pm f | \mathbf{1}_{\mathcal{F}_\pm g} \rangle. \end{aligned}$$

To prove that $\mathcal{F}_\pm \mathcal{F}_\pm^* = \mathbf{1}_{\mathcal{K}}$, it therefore suffices to show that $\mathcal{F}_\pm(\ell^2(\mathbb{Z}, \mathbb{C}^L))$ is a dense subspace of $\mathcal{K}$, which is equivalent to $\mathcal{F}_\pm(\ell^2(\mathbb{Z}, \mathbb{C}))^\perp = \{0\}$.

So, let $f \in \mathcal{F}_\pm(\ell^2(\mathbb{Z}, \mathbb{C}))^\perp$. Then, for every interval $I \subseteq \mathbb{R}$ and every $g \in \ell^2(\mathbb{Z}, \mathbb{C})$, one has

$$\begin{aligned} 0 &= \langle \mathcal{F}_\pm P_I(H)g | f \rangle = \langle \chi_I \mathcal{F}_\pm g | f \rangle \\ &= \sum_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{[-2a_k, 2a_k]} \chi_I(E) \overline{(\mathcal{F}_\pm g)_{k, \sigma}(E)} f_{k, \sigma}(E) dE \\ &= \int_I \sum_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \chi_{[-2a_k, 2a_k]}(E) \overline{(\mathcal{F}_\pm g)_{k, \sigma}(E)} f_{k, \sigma}(E) dE, \end{aligned}$$

using (b) for the second equality. Therefore, for every $g \in \ell^2(\mathbb{Z}, \mathbb{C})$, one has

$$0 = \sum_{\substack{k \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \chi_{[-2a_k, 2a_k]}(E) \overline{(\mathcal{F}_\pm g)_{k, \sigma}(E)} f_{k, \sigma}(E) = \sum_{\substack{k \in J_E \\ \sigma \in \{+, -\}}} \overline{(\mathcal{F}_\pm g)_{k, \sigma}(E)} f_{k, \sigma}(E)$$

for almost every $E \in \mathbb{R}$, using Proposition ?? for the last equality. We now consider specific functions of the form $g = \delta_n e_l$ with $n \in \mathbb{Z}$ and $l \in \{1, \dots, L\}$. Using that

$$\overline{(\mathcal{F}_\pm(\delta_n e_l))_{k, \sigma}(E)} = \overline{\langle w_{k, \pm}^{E, \sigma}(n) | e_l \rangle} = \langle e_l | w_{k, \pm}^{E, \sigma}(n) \rangle,$$

for every $n \in \mathbb{Z}$ and $l \in \{1, \dots, L\}$, we get

$$0 = \sum_{\substack{k \in J_E \\ \sigma \in \{+, -\}}} \langle e_l | w_{k, \pm}^{E, \sigma}(n) \rangle f_{k, \sigma}(E)$$

for almost every $E \in \mathbb{R}$. Since the set

$$\{(n, l) : n \in \mathbb{Z}, l \in \{1, \dots, L\}\}$$

is countable, this implies that, for almost every $E \in \mathbb{R}$, one has

$$0 = \sum_{\substack{k \in J_E \\ \sigma \in \{+, -\}}} \langle e_l | w_{k, \pm}^{E, \sigma}(n) \rangle f_{k, \sigma}(E) = \left\langle e_l \left| \sum_{\substack{k \in J_E \\ \sigma \in \{+, -\}}} f_{k, \sigma}(E) w_{k, \pm}^{E, \sigma}(n) \right. \right\rangle$$

for every $n \in \mathbb{Z}$ and $l \in \{1, \dots, L\}$ and therefore

$$0 = \sum_{\substack{k \in J_E \\ \sigma \in \{+, -\}}} f_{k, \sigma}(E) w_{k, \pm}^{E, \sigma}(n) \quad \forall n \in \mathbb{Z},$$

which implies

$$0 = \sum_{\substack{k \in J_E \\ \sigma \in \{+, -\}}} f_{k, \sigma}(E) w_{k, \pm}^{E, \sigma}.$$

Since

$$\left\{ w_{k, \pm}^{E, \sigma} : k \in J_E, \sigma \in \{+, -\} \right\}$$

forms a basis of $\mathcal{B}(H, E)$, this implies that, for almost every $E \in \mathbb{R}$, one has

$$f_{k,\sigma}(E) = 0 \quad \forall k \in J_E, \sigma \in \{+, -\}$$

and therefore $f = 0$. $\square$

We finish this section with a result that we will need later to calculate the wave operators:

Proposition 2.2.6. *Let $f \in \mathcal{K}$, $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$ and $E \in [-2a_1, 2a_1] \setminus \mathfrak{D}$. Then, for every null-sequence $(\varepsilon_n)_{n \in \mathbb{N}}$, there exists a subsequence $(\varepsilon_{n_l})_{l \in \mathbb{N}}$ such that*

$$\lim_{l \rightarrow \infty} \int^{\mathcal{K}} \overline{f(E)} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_l})^{-1} g)(E) dE = \int^{\mathcal{K}} \overline{f(E)} ((\mathcal{F}_0 - \mathcal{F}_{\pm})g)(E) dE.$$

Proof. By Theorem ??(g) and Lemma ??(b), the sequence $((H_0 - E \pm i\varepsilon_n)^{-1} g)_{n \in \mathbb{N}}$ is bounded in $\ell^\infty(\mathbb{Z}, \mathbb{C}^L)$ and therefore, the sequence $(V(H_0 - E \pm i\varepsilon_n)^{-1} g)_{n \in \mathbb{N}}$ is bounded in $\ell^1(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$. Then, using that, by Theorem 2.2.5, $\mathcal{F}_{\pm}$ is a bounded operator, also $(\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_n)^{-1} g)_{n \in \mathbb{N}}$ is a bounded sequence in $\mathcal{K}$. By the Banach-Alaoglu Theorem it then has a weakly convergent subsequence $(\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_l})^{-1} g)_{l \in \mathbb{N}}$, i.e. there exists a function $G \in \mathcal{K}$, such that

$$\lim_{l \rightarrow \infty} \int^{\mathcal{K}} \overline{f(E)} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_l})^{-1} g)(E) dE = \int^{\mathcal{K}} \overline{f(E)} G(E) dE.$$

This implies that there exists a subsequence $(\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_{l_m}})^{-1} g)_{m \in \mathbb{N}}$ that converges almost everywhere pointwise to G , i.e. for every $k \in \{1, \dots, L\}$ and $\sigma \in \{\pm\}$, one has

$$G_{k,\sigma}(E) = \lim_{m \rightarrow \infty} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_{l_m}})^{-1} g)_{k,\sigma}(E)$$

for almost every $E \in [-2a_k, 2a_k]$. The statement then follows since, by Theorem ??, for every $E \in [-2a_k, 2a_k] \setminus \mathfrak{D}$, one has

$$\begin{aligned} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon)^{-1} g)_{k,\sigma}(E) &= \sum_{n \in \mathbb{Z}} \left\langle w_{k,\pm}^{E,\sigma}(n) \middle| V(H_0 - E \pm i\varepsilon)^{-1} g(n) \right\rangle_{\mathbb{C}^L} \\ &= \sum_{n \in \mathbb{Z}} \left\langle (H_0 - E \mp i\varepsilon)^{-1} V w_{k,\pm}^{E,\sigma}(n) \middle| g(n) \right\rangle_{\mathbb{C}^L} \\ &\xrightarrow{\varepsilon \downarrow 0} \sum_{n \in \mathbb{Z}} \left\langle v_k^{E,\sigma}(n) - w_{k,\pm}^{E,\sigma}(n) \middle| g(n) \right\rangle_{\mathbb{C}^L} \\ &= ((\mathcal{F}_0 - \mathcal{F}_{\pm})g)_{k,\sigma}(E). \end{aligned} \quad \square$$

3 The wave- and scattering-operators

In this section, we summarize our previous results in calculating the wave and scattering operators. We start with the wave operators:

Theorem 3.1. *One has $\Omega_{\pm} = \mathcal{F}_{\pm}^* \mathcal{F}_0$.*

Proof. We have

$$\frac{d}{ds} e^{isH} e^{-isH_0} = e^{isH} (iH - iH_0) e^{-isH_0} = ie^{isH} V e^{-isH_0},$$

so

$$e^{itH} e^{-itH_0} = e^{i0H} e^{-i0H_0} + \int_0^t \frac{d}{ds} e^{isH} e^{-isH_0} ds = \mathbf{1} + i \int_0^t e^{isH} V e^{-isH_0} ds.$$

For $f, g \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$, using that, by Proposition 1.1 and Theorem 2.2.5, we have

$$\Omega_{\pm} = P_{\text{ac}}(H) \Omega_{\pm} = \mathcal{F}_{\pm}^* \mathcal{F}_{\pm} \Omega_{\pm},$$

this yields

$$\begin{aligned} \langle f | \Omega_{\pm} g \rangle &= \langle f | \mathcal{F}_{\pm}^* \mathcal{F}_{\pm} \Omega_{\pm} g \rangle = \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} \Omega_{\pm} g \rangle \\ &= \lim_{t \rightarrow \mp\infty} \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} e^{itH} e^{-itH_0} g \rangle \\ &= \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} g \rangle + i \lim_{t \rightarrow \mp\infty} \int_0^t \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} e^{isH} V e^{-isH_0} g \rangle ds \\ &= \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} g \rangle + i \lim_{\varepsilon \downarrow 0} \int_0^{\mp\infty} \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} e^{isH} V e^{-isH_0} g \rangle e^{\pm\varepsilon s} ds, \end{aligned} \quad (15)$$

using [RS79, Sec. XI.6: Lem. 5] in the last step. Further, using that, by Theorem 2.2.5(b), we have

$$\mathcal{F}_{\pm} e^{isH} = e^{isM_{\text{Id}}} \mathcal{F}_{\pm},$$

for the integrand, we get

$$\begin{aligned} \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} e^{isH} V e^{-isH_0} g \rangle e^{\pm\varepsilon s} &= \langle \mathcal{F}_{\pm} f | e^{isM_{\text{Id}}} \mathcal{F}_{\pm} V e^{-isH_0} g \rangle e^{\pm\varepsilon s} \\ &= \int^{\mathcal{K}} \overline{(\mathcal{F}_{\pm} f)(E)} e^{isE} (\mathcal{F}_{\pm} V e^{-isH_0} g)(E) e^{\pm\varepsilon s} dE \\ &= \int^{\mathcal{K}} \overline{(\mathcal{F}_{\pm} f)(E)} (\mathcal{F}_{\pm} V e^{-is(H_0 - E \pm i\varepsilon)} g)(E) dE. \end{aligned}$$

Interchanging integrals with the Fubini–Tonelli Theorem, this yields

$$\begin{aligned} &i \int_0^{\mp\infty} \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} e^{isH} V e^{-isH_0} g \rangle e^{\pm\varepsilon s} ds \\ &= \int^{\mathcal{K}} \overline{(\mathcal{F}_{\pm} f)(E)} \left(\mathcal{F}_{\pm} V \int_0^{\mp\infty} i e^{-is(H_0 - E \pm i\varepsilon)} ds g \right) (E) dE \\ &= \int^{\mathcal{K}} \overline{(\mathcal{F}_{\pm} f)(E)} (\mathcal{F}_{\pm} V (H_0 - E \pm i\varepsilon)^{-1} g) (E) dE. \end{aligned} \quad (16)$$

Now, by Proposition 2.2.6, for $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, there exists a null-sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that

$$\begin{aligned} &\int^{\mathcal{K}} \overline{(\mathcal{F}_{\pm} f)(E)} (\mathcal{F}_{\pm} V (H_0 - E \pm i\varepsilon_n)^{-1} g) (E) dE \\ &\xrightarrow{n \rightarrow \infty} \int^{\mathcal{K}} \overline{(\mathcal{F}_{\pm} f)(E)} ((\mathcal{F}_0 - \mathcal{F}_{\pm})g)(E) dE = \langle \mathcal{F}_{\pm} f | (\mathcal{F}_0 - \mathcal{F}_{\pm})g \rangle. \end{aligned}$$

Together with Equations (15) and (16), we then get

$$\begin{aligned}
\langle f | \Omega_{\pm} g \rangle &= \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} g \rangle + \lim_{\varepsilon \downarrow 0} i \int_0^{\mp \infty} \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} e^{isH} V e^{-isH_0} g \rangle e^{\pm \varepsilon s} ds \\
&= \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} g \rangle + \lim_{n \rightarrow \infty} \int_{\mathcal{K}} \overline{(\mathcal{F}_{\pm} f)(E)} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_n)^{-1} g)(E) dE \\
&= \langle \mathcal{F}_{\pm} f | \mathcal{F}_{\pm} g \rangle + \langle \mathcal{F}_{\pm} f | (\mathcal{F}_0 - \mathcal{F}_{\pm}) g \rangle = \langle \mathcal{F}_{\pm} f | \mathcal{F}_0 g \rangle = \langle f | \mathcal{F}_{\pm}^* \mathcal{F}_0 g \rangle.
\end{aligned}$$

We therefore have

$$\langle f | \Omega_{\pm} g \rangle = \langle f | \mathcal{F}_{\pm}^* \mathcal{F}_0 g \rangle$$

for every $f \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$ and $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, which yields $\Omega_{\pm} = \mathcal{F}_{\pm}^* \mathcal{F}_0$ since $c_{00}(\mathbb{Z}, \mathbb{C}^L)$ is dense in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$. $\square$

We now want to calculate the scattering operator. For this we need the scattering matrix introduced in Definition ???. By definition, we have

$$S^E \in \mathcal{M}_{2|J^E| \times 2|J^E|}(\mathbb{C}), \quad E \in [-2a_1, 2a_1] \setminus \mathfrak{D}.$$

Since, by Proposition ??, $\mathfrak{D}$ is a Lebesgue zero-set and the matrix S^E is unitary for every $E \in [-2a_1, 2a_1] \setminus \mathfrak{D} = \sigma(H_0) \setminus \mathfrak{D}$, taking direct integrals, we get a unitary operator

$$\int_{\sigma(H_0)}^{\oplus} S^E dE \in \mathcal{U} \left(\int_{\sigma(H_0)}^{\oplus} \mathbb{C}^{2|J^E|} dE \right).$$

Now, defining

$$I_k := \begin{cases} (-2a_k, -2a_{k+1}) \cup (2a_{k+1}, 2a_k) & \text{if } k = 1, \dots, L-1 \\ (-2a_L, 2a_L) & \text{if } k = L, \end{cases}$$

by Proposition ??, for $E \in I_k$, we have $|J^E| = k$. This yields

$$\begin{aligned}
\int_{\sigma(H_0)}^{\oplus} \mathbb{C}^{2|J^E|} dE &\cong \bigoplus_{k=1}^L L^2(I_k, \mathbb{C}^{2k}) \cong \bigoplus_{k=1}^L \bigoplus_{\substack{j \in \{1, \dots, k\} \\ \sigma \in \{+, -\}}} L^2(I_k, \mathbb{C}) = \bigoplus_{\substack{j \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \bigoplus_{k=j}^L L^2(I_k, \mathbb{C}) \\
&\cong \bigoplus_{\substack{j \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} L^2 \left(\bigcup_{k=l}^L I_k, \mathbb{C} \right) \cong \bigoplus_{\substack{j \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} L^2([-2a_j, 2a_j], \mathbb{C}) = \mathcal{K}, \quad (17)
\end{aligned}$$

so we can interpret $\int_{\sigma(H_0)}^{\oplus} S^E dE$ as a unitary operator on $\mathcal{K}$. We will now show that this operator is unitarily equivalent to the scattering operator:

Theorem 3.2. *The following hold:*

- (a) $\mathcal{F}_- = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_+.$
- (b) $S = \mathcal{F}_0^* \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_0.$

Proof. (a) By Lemma ??, we have

$$\left(w_{k,+}^{E,\sigma} \right)_{\substack{k \in J_E \\ \sigma \in \{+,-\}}} = \left(w_{k,-}^{E,\sigma} \right)_{\substack{k \in J_E \\ \sigma \in \{+,-\}}} S^E.$$

Taking adjoints, this yields

$$\left(\left\langle w_{k,+}^{E,\sigma} \right| \right)_{\substack{k \in J_E \\ \sigma \in \{+,-\}}} = (S^E)^* \left(\left\langle w_{k,-}^{E,\sigma} \right| \right)_{\substack{k \in J_E \\ \sigma \in \{+,-\}}},$$

which implies

$$\mathcal{F}_+ = \left(\int_{\sigma(H_0)}^{\oplus} (S^E)^* dE \right) \mathcal{F}_-$$

and therefore

$$\mathcal{F}_- = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_+.$$

(b) By (a) and Theorem 2.2.5(c), we have

$$\mathcal{F}_- \mathcal{F}_+^* = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_+ \mathcal{F}_+^* = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right)$$

and therefore, by Theorem 3.1, we get

$$S = \Omega_-^* \Omega_+ = \mathcal{F}_0^* \mathcal{F}_- \mathcal{F}_+^* \mathcal{F}_0 = \mathcal{F}_0^* \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_0. \quad \square$$

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